BIOINFORMATICS II · FINAL RESEARCH PAPER

Data-Driven Hyperparameter Optimization for Accurate Sparse Deconvolution of Spatial Transcriptomic Data

Prepared by: Habiba Hammouda

THE BIG IDEA · IN PLAIN TERMS

Think of It Like Un-Blending a Smoothie

A tissue spot is like a sip of fruit smoothie.

You can measure the sip, but the individual fruits are blended together and hidden. The goal is to work out which fruits are inside — and how much of each.

THE SIP

A blended sip of smoothie

↓ is really

A 55 µm Visium spot holding 1–10 cells, whose signals are mixed into one reading.

THE TASTE-MEMORY

Knowing pure banana, pure strawberry…

↓ is really

A single-cell RNA reference — the gene 'fingerprint' of each of the 26 pure cell types.

THE DETECTIVE

Un-blending to name the fruits & amounts

↓ is really

WISpR deconvolution — estimating which cell types are present and in what proportion.

THE QUESTION whether the detective must re-learn its technique for **every single sip** — or whether one fixed technique would do.

One Simple Question

WISpR uses a slow, careful tuning step (GridSearch) to pick its settings separately for every spot in the tissue.

Is that slow step actually necessary — or could simple, fixed settings work just as well?

1

Set up
What the problem is, what WISpR does, and how it was tested

2

Results
How the careful baseline performed, validated against real biology

3

The answer
What happens when simple fixed settings are used instead

Why This Question Matters

1

GridSearch is slow

Tuning every spot took ~32 minutes for 2,110 spots. On larger datasets this becomes a real bottleneck.

2

Shortcuts are tempting

In practice, some users substitute fixed, manually-chosen settings to save time. Is that safe?

3

Never tested for WISpR

No one had systematically checked whether spot-specific tuning is truly necessary. This project does.

Each Measurement is a Mix of Cells

THE TECHNOLOGY

Spatial transcriptomics (10X Visium) measures which genes are active in tissue, while keeping track of WHERE each measurement came from.

THE CATCH

Each spot is **55 µm wide** and captures **1–10 cells** whose signals are blended together. The individual cell types are hidden.

THE SOLUTION: DECONVOLUTION

Use single-cell RNA data as a reference to un-mix each spot — estimating which cell types are present and in what proportion.

$$y = X \cdot \beta$$

y = spot's measured genes X = reference signatures
β = cell-type proportions (the unknown)

WISpR — Weight-Induced Sparse Regression

Erdogan & Eroglu (2025) · the method this project tests

Biological sparsity

Forces each spot to contain only a few cell types — because a tiny spot physically can't hold many cells.

Spot-specific tuning

Optimizes its settings SEPARATELY for every single spot, not one fixed setting for all. (This is what is tested.)

Iterative thresholding

Repeatedly prunes weak predictions (SINDy-inspired) until only well-supported cell types remain.

TWO HYPERPARAMETERS ("KNOBS") CONTROL EVERYTHING:

τ (tau) how aggressively weak cell types are removed

λ (lambda) regression stability (L2 regularization)

The Experimental Design

DATASET Mouse hippocampus · 10X Visium · **2,110 spots** · **26 cell types** · Reference: Linnarsson Mouse Brain Atlas

A fair contest: GridSearch baseline vs. four fixed manual settings

BASELINE	MANUAL 1	MANUAL 2	MANUAL 3	MANUAL 4
GridSearch	**Loose threshold**	**Strict threshold**	**Weak penalty**	**Strong penalty**
tuned per spot	$\tau=0.05, \lambda=10$	$\tau=0.30, \lambda=10$	$\tau=0.15, \lambda=5$	$\tau=0.15, \lambda=50$

HYPOTHESIS If spot-specific tuning is truly necessary, all four fixed manual settings should produce worse predictions — higher error and biologically impossible cell counts. Measured by **RMSE** (error) and **sparsity** (cell types per spot).

The Analysis Pipeline

1 Preprocess

Normalize to 10k counts, log1p; top 2,000 variable genes (Scanpy).

2 Build reference

26 cell types from 1,754 cells; clusters kept at 25–250 cells each.

3 Select markers

Gini-coefficient DEGs (≥0.30); 2,000-gene reference matrix, mean Gini 0.857.

4 Deconvolve

Non-negative regularized regression + iterative soft-thresholding, per spot.

5 Optimize

GridSearchCV: $\tau \in [0.001, 0.500]$, $\lambda \in \{0.001 \ldots 10\}$, 5-fold CV ×3.

6 Evaluate

RMSE, sparsity, marker correlations; Wilcoxon tests + effect sizes.

The Baseline Works Beautifully

GRIDSEARCH BASELINE

4.22

cell types per spot (mean)

Biologically realistic — a 55 µm spot holds only a handful of cells.

0.454

median reconstruction error (RMSE)

Low error — predictions reconstruct the measured data closely.

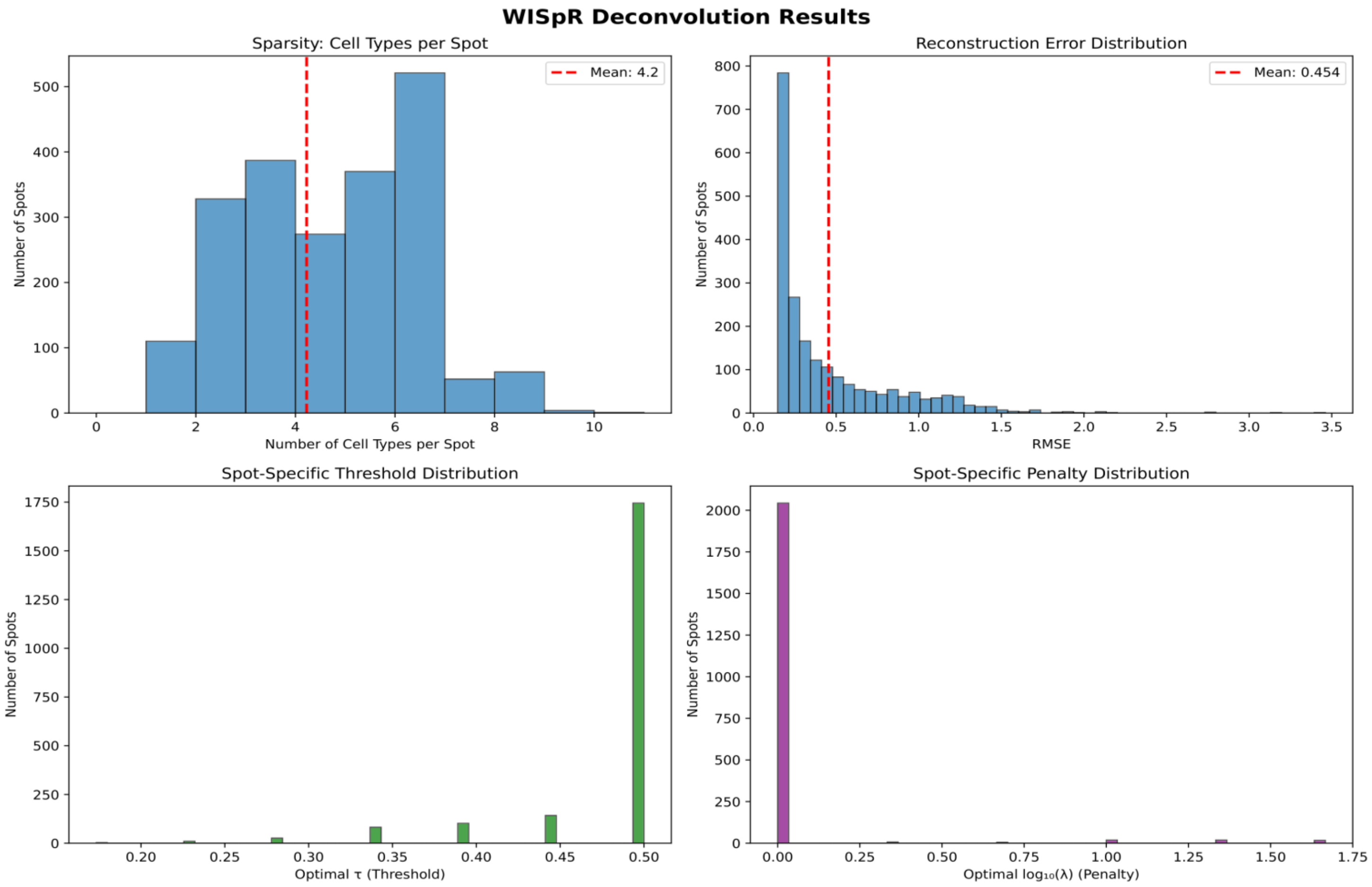
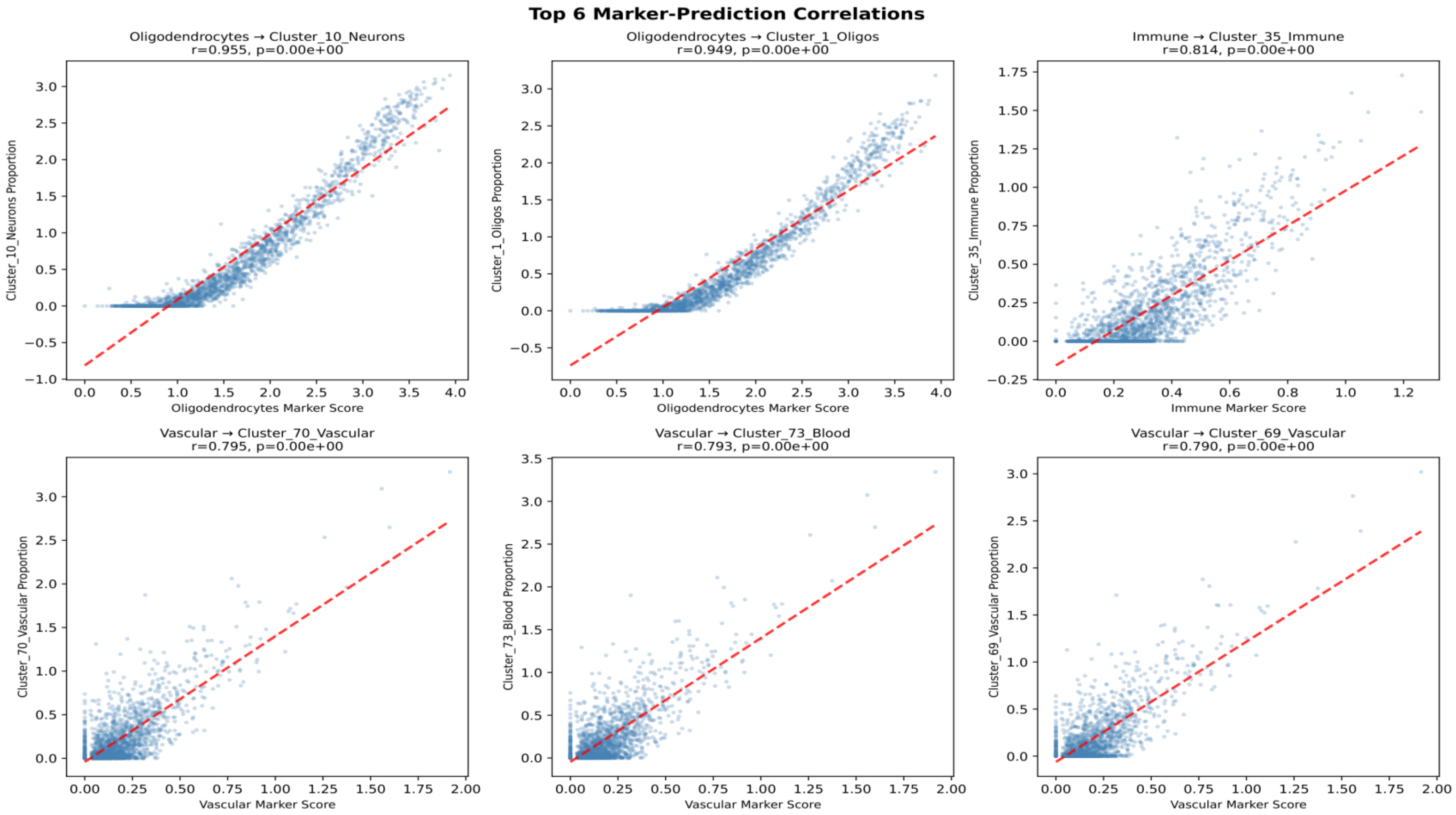


Figure 1 — Sparsity, error, and the tuned settings (τ peaks at ≈0.50; λ ≈ 1.0).

Predictions Match Real Biology

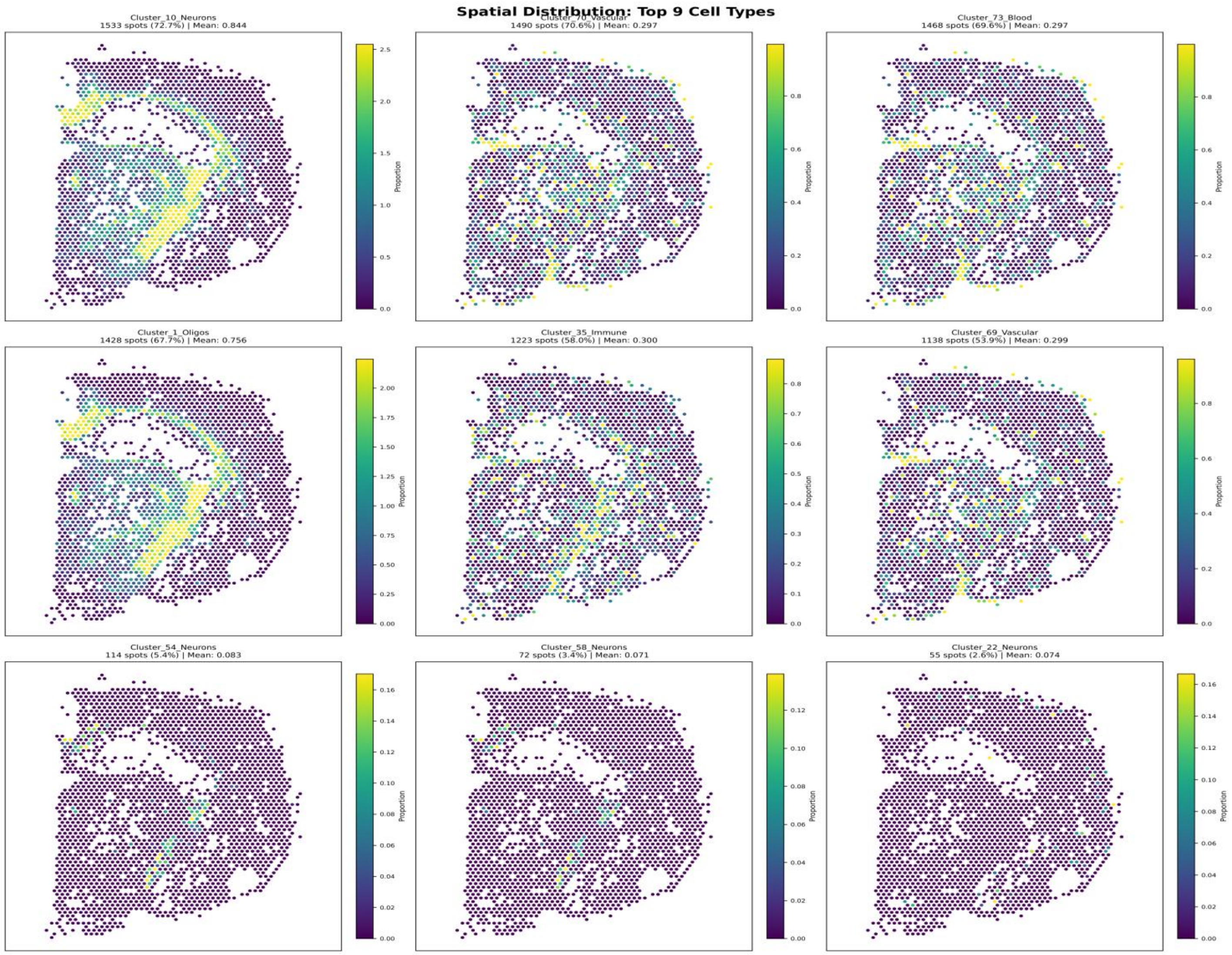
Validated independently against canonical marker genes

r = 0.95	Oligodendrocyte markers
r = 0.81	Immune markers
r = 0.79	Vascular markers
$p < 10^{-230}$	all correlations



Note: a cluster labeled "Neurons" in the reference correlated with oligodendrocyte markers (r=0.955) — the validation caught a mislabel in the atlas.

Spatial Patterns Match Anatomy



WHAT THE MAPS SHOW

- **Oligodendrocytes**
  concentrate centrally — white-matter regions
- **Vascular cells**
  scattered, network-like — blood vessels
- **Immune cells**
  diffuse coverage — surveillance microglia
- **Rare neurons**
  specific zones only (2.6–5.4% of spots)

Real biology — not random noise.

What Happens With Fixed Settings?

✓ GRIDSEARCH BASELINE

4.22

cell types per spot

RMSE = 0.45

✓ Biologically realistic
✓ Low reconstruction error
✓ Validated by marker genes

✗ FIXED MANUAL SETTINGS

12–26

cell types per spot (impossible!)

RMSE = 0.78 – 10.51

✗ 41.6%–95.7% higher error
✗ Biologically impossible
✗ All $p < 10^{-100}$

Every Fixed Setting Fails — Significantly

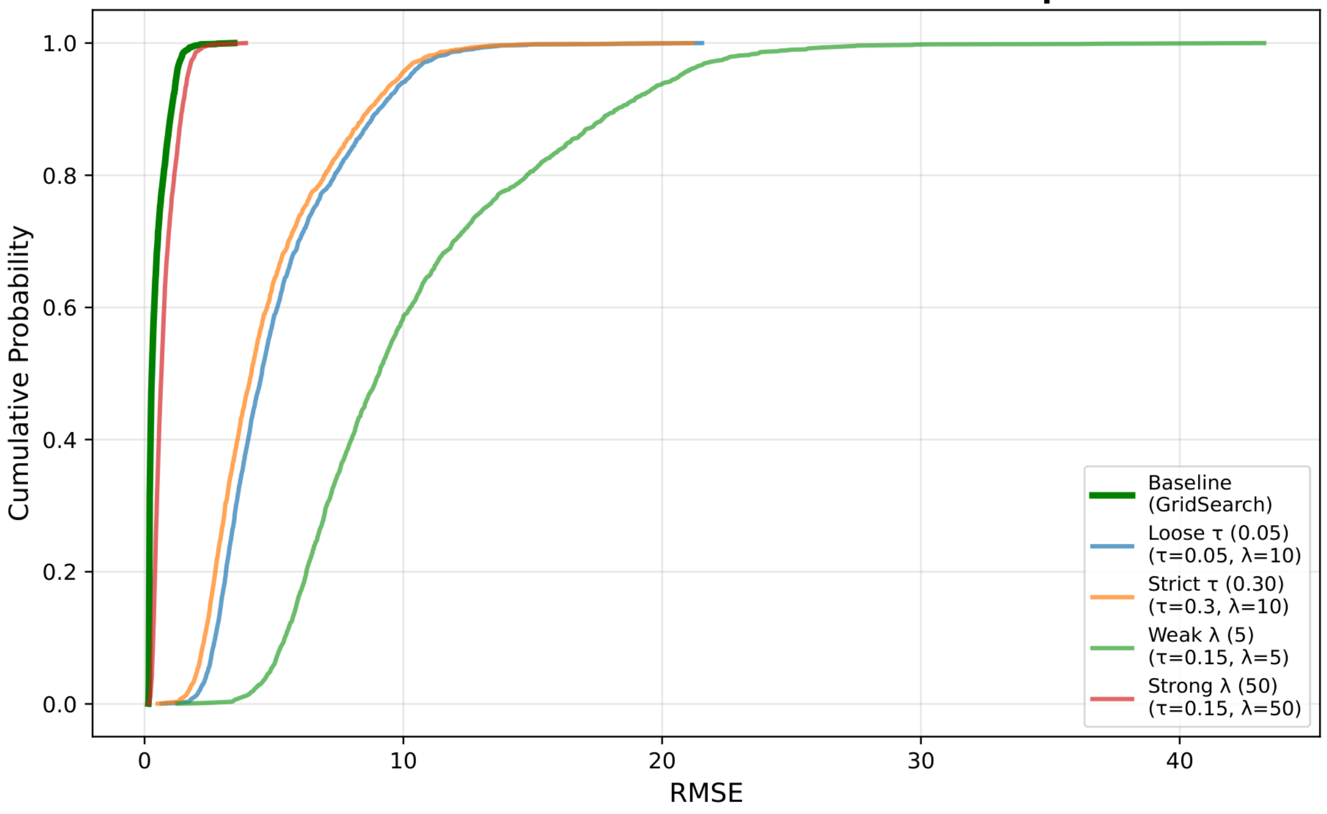


Figure 4 — Per-spot RMSE (CDF). Baseline (green) is far left = lowest error.

EFFECT SIZES (Cohen's d)

Loose τ	RMSE +91.3%	*d = −2.64*
Strict τ	RMSE +90.6%	*d = −2.41*
Weak λ	RMSE +95.7%	*d = −2.76*
Strong λ	RMSE +41.6%	*d = −0.77*

All $p < 10^{-100}$

Cliff's δ ≈ −1.0 → baseline wins on nearly every spot

Interpretation – Why Do Fixed Settings Fail?

Tissue is spatially heterogeneous — every region is different (every smoothie sip is different)

White matter, grey matter, vasculature, and tissue edges all have different cell compositions, signal-to-noise, and expression variability. One fixed (τ, λ) cannot fit them all.

GridSearch adapts

Tunes each spot to its local context, so it stays accurate everywhere across the tissue.

Telling detail: 82.7% of spots pushed τ to the grid's upper bound (≈0.50) — aggressive sparsity is genuinely needed.

Fixed settings can't

A single setting is right for some regions and wrong for most — so it either keeps every cell type (no sparsity) or destabilizes and overfits noise.

Result: 3–6× too many cell types per spot, and far higher error.

Key Takeaways

1 **Spot-specific GridSearch is essential — not optional**
Every fixed configuration failed significantly (41–96% higher error).

2 **Tissue heterogeneity defeats uniform settings**
Different regions need different parameters; one-size-fits-all can't work biologically.

3 **Marker-gene validation is powerful**
Strong correlations ($r > 0.8$) confirm predictions are biology, not artifacts — and caught a reference mislabel.

4 **The 32-minute cost is worth it**
It is the difference between biologically meaningful results and impossible ones.

Limitations & Future Directions

⚠ LIMITATIONS (TA's comments)

Single tissue

Only mouse hippocampus tested.

Matched reference

No cross-species / mismatched data.

Grid boundary

Most spots hit upper τ — a wider range may help.

No method comparison

WISpR only, not vs. RCTD / Cell2location.

→ FUTURE WORK

More tissues

Brain, heart, tumor microenvironment.

Faster tuning

Bayesian optimization, surrogate models.

Visium-HD

Scale to subcellular resolution.

Mismatch robustness

Cross-species / cross-tissue references.

LAST NOTES ON RESEARCH & PhD

sometimes "no output" is an output

more hours ≠ more achievement

Greater appreciation for researchers now!

idea changes with every run

AI is not useful as I hoped

results are not always promising

2 months… not enough

Thank You

Any questions?